Chassis Design Notes — Bottom Plate v2

Overview

Differential-drive base plate, designed parametrically in Fusion 360, printed in PETG on a Bambu Lab P1S.

Dimensions

- Plate: 150 × 120 × 3 mm, R6 rounded corners
- Coordinates below are measured from the exported STL, origin at the plate corner (0, 0)

Features (verified from the STL)

Feature	**Qty**	**Ø**	**Positions (X, Y)**
Motor mount holes (zip-tie / M3)	8	3.2 mm	Left: (5,30) (5,47) (25,30) (25,47) · Right: (125,30) (125,47) (145,30) (145,47)
Second-deck standoff holes (M3)	4	3.5 mm	(20,15) (130,15) (20,105) (130,105) → 110 × 90 mm pattern
Caster mount holes	2	4.2 mm	(55,100) (95,100) → 40 mm spacing

Drive axle line ≈ Y 38.5 (rear) · caster at Y ≈ 100 (front) → wheelbase ≈ 61.5 mm.

v1 → v2 changes

- v1 (140 × 200 mm) had misaligned motor holes and excess length. v2 resized to 150 × 120 mm and re-dimensioned from actual motor measurements.
- R6 corner fillets added; 4-hole standoff pattern added for a future second deck (Raspberry Pi + electronics).

Mounting decisions

- TT motors sit on top of the plate, held by zip ties through the Ø3.2 holes. Watch item: if the ties loosen under vibration, switch to printed L-brackets (planned fallback).
- Ball caster bolts under the front. With the motors on top, the plate bottom sits ≈ 18–19 mm above ground (66 mm wheels, axle ≈ 11 mm above the plate), which matches the caster height — no riser needed.
- The full-size breadboard (165 × 55 mm) is intentionally NOT carried on the robot — it stays on the bench for prototyping. On-robot wiring uses direct jumpers; a mini breadboard will carry the ECHO voltage divider + IMU from the STM32 phase onward.

Print settings

PETG · 0.2 mm layers · 3 walls · 25–30 % infill · printed flat, largest face down (no supports needed)